

<HLS Programming with FPGAs>

Amazon Web Service
(AWS) F1

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Acknowledgement

- Many of the figures and contents in this part are from
 - <https://github.com/aws/aws-fpga>
 - Announcing Amazon EC2 F1 Instances with Custom FPGA (<https://youtu.be/5sDVprory3c>)
 - Amazon EC2 F1 SDAccel Developer Lab (<https://youtu.be/RvTSyVa6bCw>)
 - Amazon EC2 Foundation (AWS re: Invent)
 - <https://aws.amazon.com/ec2>

Learning Objectives

- Explain the architecture of AWS F1 instance
- Create an AWS instance and log into the instance

Amazon Web Service

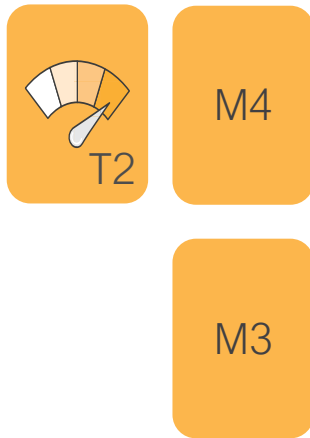


- On-demand cloud computing service provided by Amazon
- Pay-as-you-go basis
- Can be used for machine learning, high-performance computing, application service, application development, etc
- Leading cloud service provider (45%** market share) as of 2019

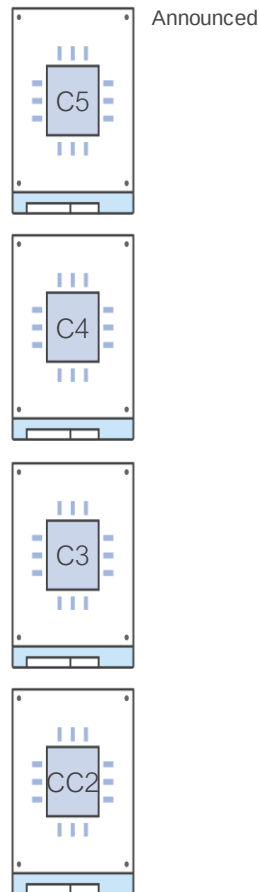
**Gartner

Amazon EC2 Instance Types

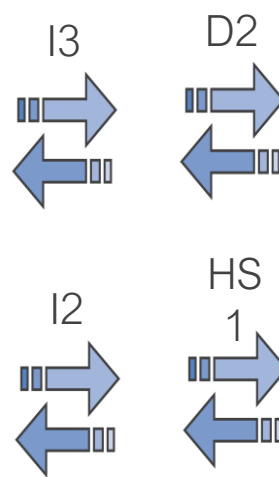
General
purpose



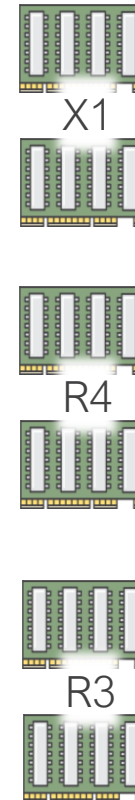
Compute
optimized



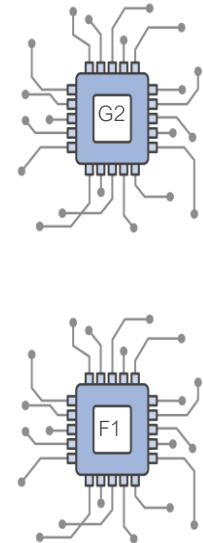
Storage and IO
optimized



Memory
optimized



GPU and FPGA
accelerated



Instance generation *Additional capabilities
M5d.xlarge
Instance family Instance size
Instance type

FPGA Accelerated Computing in AWS

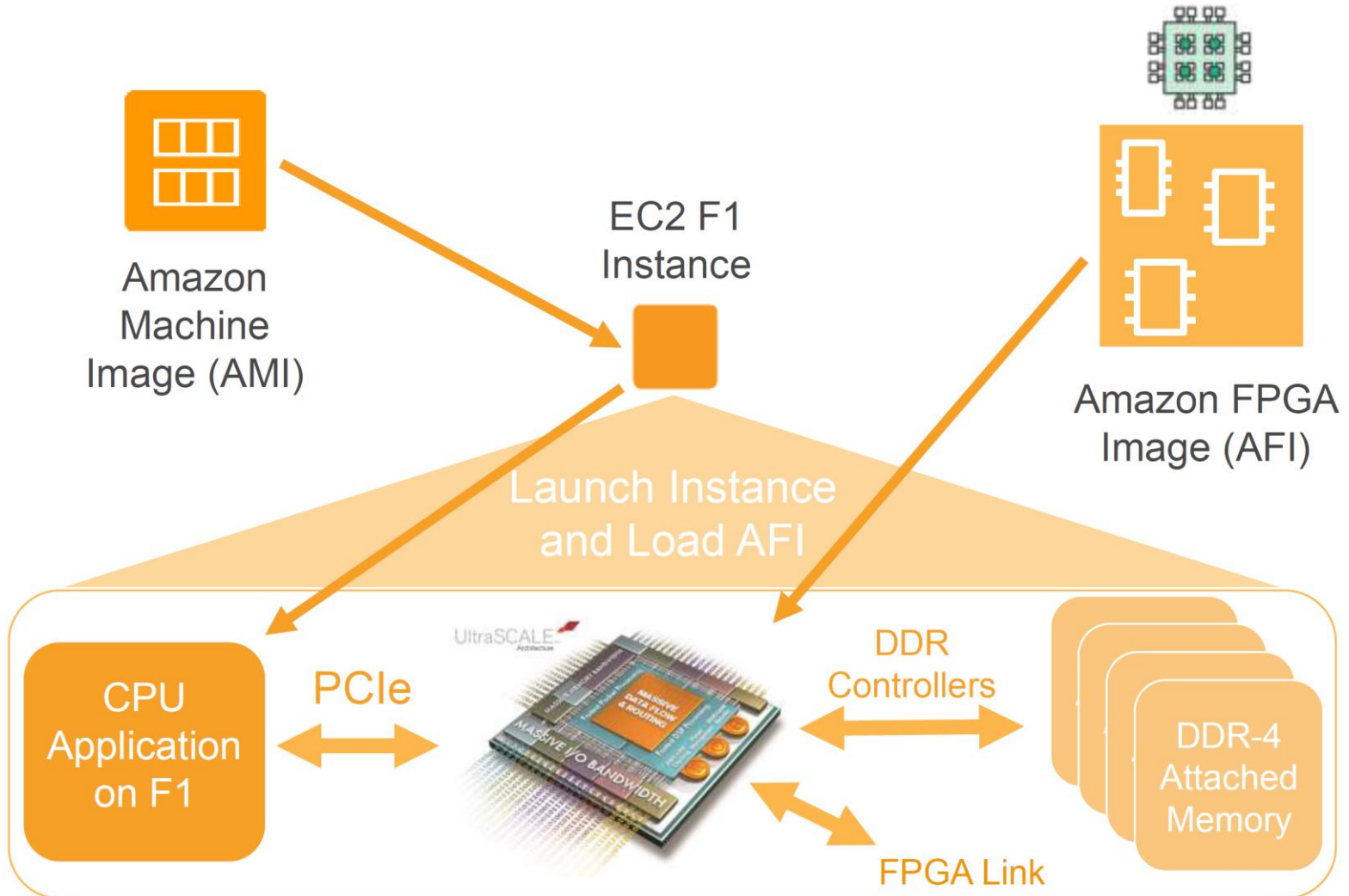
Instance	FPGAs	vCPU	Mem (GiB)	SSD Storage (GB)	Networking Performance
f1.2xlarge	1	8	122	470	Up to 10 Gigabit
f1.4xlarge	2	16	244	940	Up to 10 Gigabit
f1.16xlarge	8	64	976	4 x 940	25 Gigabit

<https://aws.amazon.com/ec2/instance-types/>

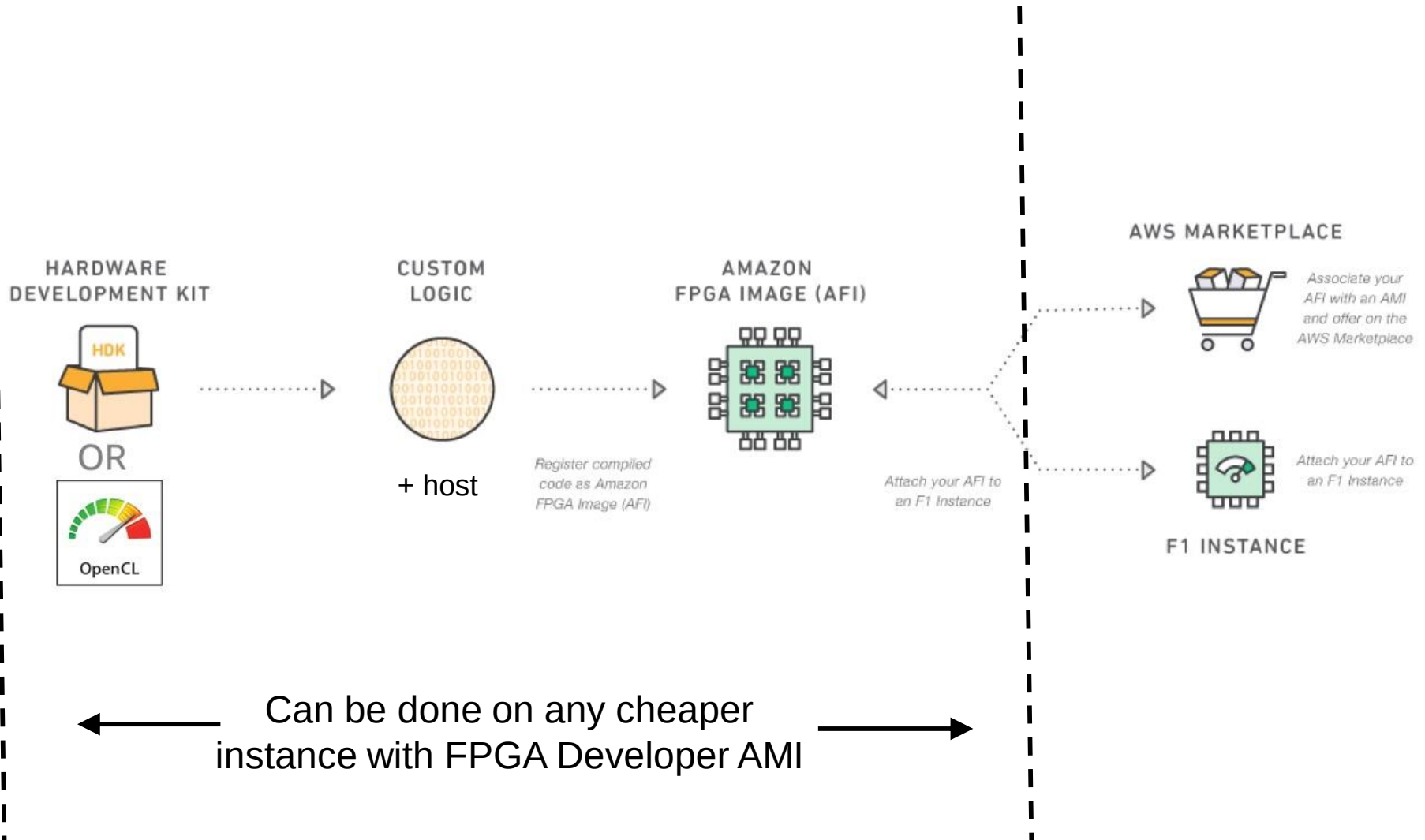


1~8 of Xilinx UltraScale+ VU9P

FPGA Acceleration Using F1



Development Flow on AWS F1



Instance Pricing

Instance name ▲	On-Demand hourly rate ▼	vCPU ▼	Memory ▼	Storage ▼	Network performance ▼
m5.large	\$0.096	2	8 GiB	EBS Only	Up to 10 Gigabit
m5.xlarge	\$0.192	4	16 GiB	EBS Only	Up to 10 Gigabit
m5.2xlarge	\$0.384	8	32 GiB	EBS Only	Up to 10 Gigabit
m5.4xlarge	\$0.768	16	64 GiB	EBS Only	Up to 10 Gigabit
m5.8xlarge	\$1.536	32	128 GiB	EBS Only	10 Gigabit

Instance name ▲	On-Demand hourly rate ▼	vCPU ▼	Memory ▼	Storage ▼	Network performance ▼
f1.2xlarge	\$1.65	8	122 GiB	1 x 470 NVMe SSD	Up to 10 Gigabit
f1.4xlarge	\$3.30	16	244 GiB	1 x 940 NVMe SSD	10 Gigabit
f1.16xlarge	\$13.20	64	976 GiB	4 x 940 NVMe SSD	20 Gigabit

<https://aws.amazon.com/ec2/pricing/on-demand/>

Storage Pricing

Volume Type	Price
General Purpose SSD (gp3) - Storage	\$0.08/GB-month
General Purpose SSD (gp3) - IOPS	3,000 IOPS free and \$0.005/provisioned IOPS-month over 3,000
General Purpose SSD (gp3) - Throughput	125 MB/s free and \$0.04/provisioned MB/s-month over 125
General Purpose SSD (gp2) Volumes	\$0.10 per GB-month of provisioned storage
Provisioned IOPS SSD (io2) - Storage	\$0.125/GB-month
Provisioned IOPS SSD (io2) - IOPS	\$0.065/provisioned IOPS-month up to 32,000 IOPS
	\$0.046/provisioned IOPS-month from 32,001 to 64,000 IOPS
	\$0.032/provisioned IOPS-month for greater than 64,000 IOPS†
Provisioned IOPS SSD (io1) Volumes	\$0.125 per GB-month of provisioned storage AND \$0.065 per provisioned IOPS-month
Throughput Optimized HDD (st1) Volumes	\$0.045 per GB-month of provisioned storage
Cold HDD (sc1) Volumes	\$0.015 per GB-month of provisioned storage

Creating AWS Account

- Sign up for AWS:
 - <https://portal.aws.amazon.com/billing/signup>



Explore Free Tier products with a new AWS account.

Sign up for AWS

Email address

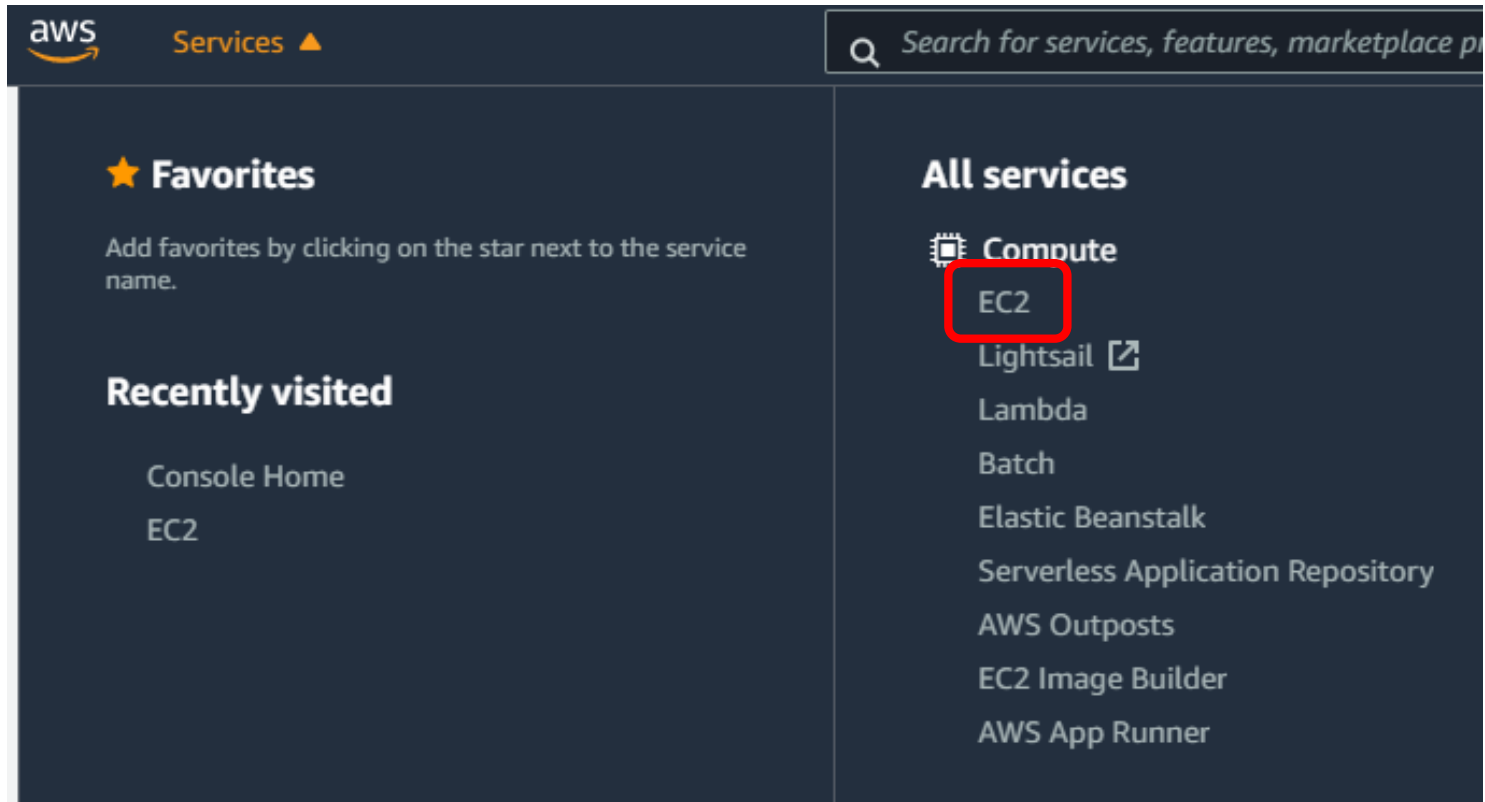
You will use this email address to sign in to your new AWS account.

AWS Promotional Credit



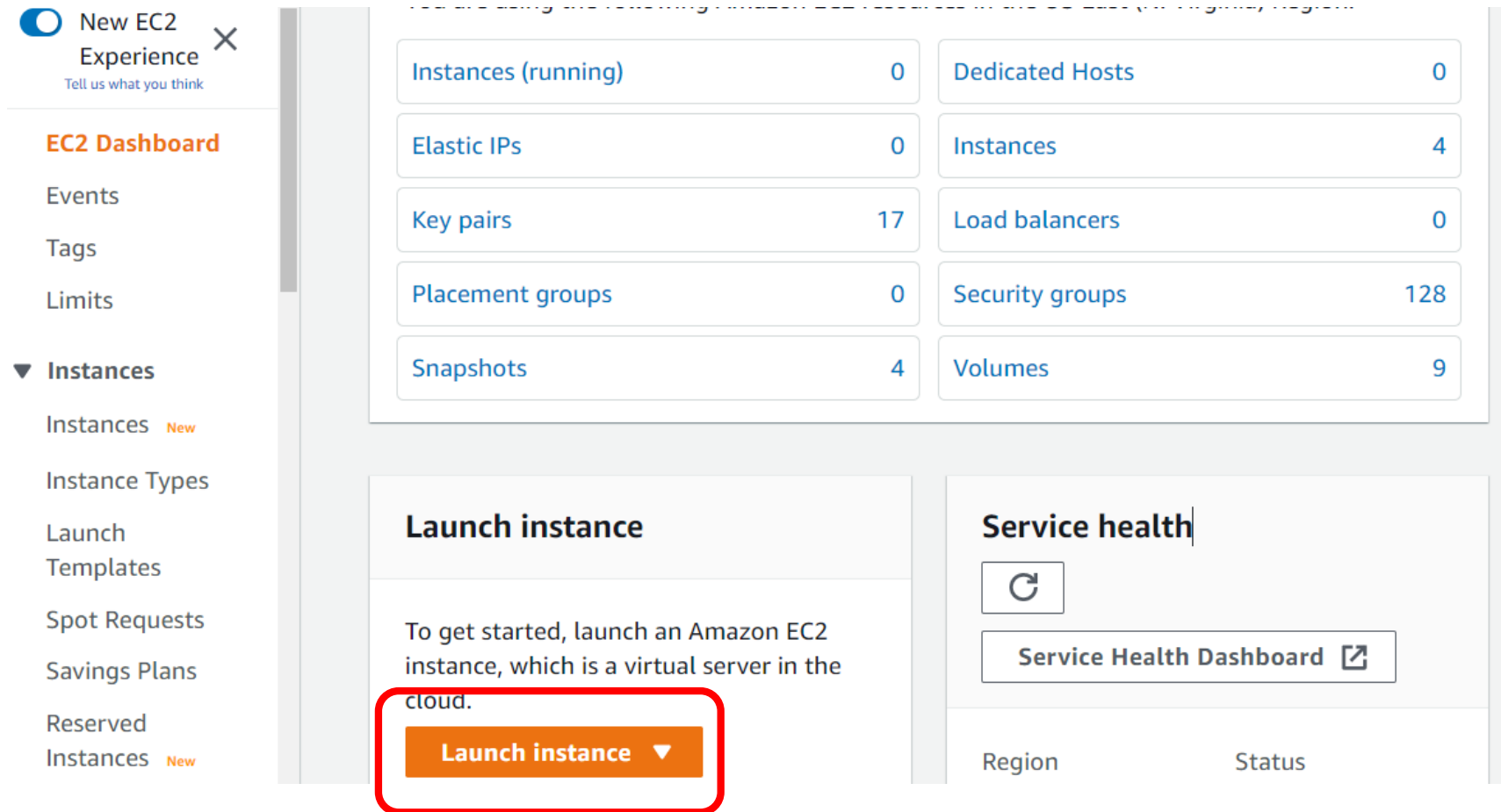
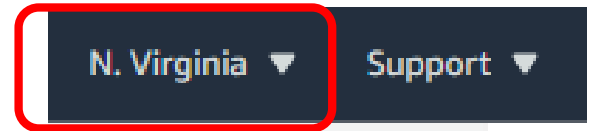
Create an AWS EC2 Instance

- Under services, click “EC2”



Create an AWS EC2 Instance

- Choose the closest region.
- Click Launch Instance under EC2 Dashboard

A screenshot of the AWS Management Console's EC2 Dashboard. On the left is a navigation sidebar with options like 'New EC2 Experience', 'EC2 Dashboard', 'Events', 'Tags', 'Limits', 'Instances' (expanded), 'Instances New', 'Instance Types', 'Launch Templates', 'Spot Requests', 'Savings Plans', 'Reserved Instances', and 'Instances New'. The main content area displays a grid of metrics: 'Instances (running)' (0), 'Elastic IPs' (0), 'Key pairs' (17), 'Placement groups' (0), 'Snapshots' (4), 'Dedicated Hosts' (0), 'Instances' (4), 'Load balancers' (0), 'Security groups' (128), and 'Volumes' (9). Below the metrics, there are two panels. The 'Launch instance' panel on the left contains the text 'To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.' and a prominent orange 'Launch instance' button with a downward arrow, which is highlighted with a red box. The 'Service health' panel on the right shows a refresh icon and a link to the 'Service Health Dashboard'.

Select AMI

Step 1: Choose an Amazon Machine Image (AMI)

[Cancel and Exit](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. You can select an AMI provided by AWS, our user community, or the AWS Marketplace; or you can select one of your own AMIs.

[Search by Systems Manager parameter](#)

Quick Start (0)

My AMIs (0)

AWS Marketplace (3)

Community AMIs (13)

Categories

All Categories

[Infrastructure Software \(3\)](#)

Architecture



Free tier eligible

FPGA Developer AMI

★★★★★ (7) | 1.11.0 [Previous versions](#) | By [Amazon Web Services](#)

Linux/Unix, CentOS 7.5 | 64-bit (x86) Amazon Machine Image (AMI) |
Updated: 7/21/21

The FPGA (field programmable gate array) AMI is a supported and maintained CentOS Linux image provided by Amazon Web Services. The AMI is pre-built with FPGA development tools and run time tools required to develop and use custom FPGAs for hardware acceleration.

[More info](#)

Select

Instance Type

Step 2: Choose an Instance Type

☐	m5	m5.large	2	8
☒	m5	m5.xlarge	4	16
☐	m5	m5.2xlarge	8	32

Instance name ▲	On-Demand hourly rate ▼	vCPU ▼	Memory ▼	Storage ▼	Network performance ▼
m5.large	\$0.096	2	8 GiB	EBS Only	Up to 10 Gigabit
m5.xlarge	\$0.192	4	16 GiB	EBS Only	Up to 10 Gigabit
m5.2xlarge	\$0.384	8	32 GiB	EBS Only	Up to 10 Gigabit
m5.4xlarge	\$0.768	16	64 GiB	EBS Only	Up to 10 Gigabit
m5.8xlarge	\$1.536	32	128 GiB	EBS Only	10 Gigabit

Create new keypair

Select an existing key pair or create a new key pair



A key pair consists of a **public key** that AWS stores, and a **private key file** that you store. Together, they allow you to connect to your instance securely. For Windows AMIs, the private key file is required to obtain the password used to log into your instance. For Linux AMIs, the private key file allows you to securely SSH into your instance.

Note: The selected key pair will be added to the set of keys authorized for this instance. Learn more about [removing existing key pairs from a public AMI](#).

Create a new key pair

Key pair name

Download Key Pair



You have to download the **private key file** (*.pem file) before you can continue. **Store it in a secure and accessible location.** You will not be able to download the file again after it's created.

Cancel

Launch Instances

Review and Launch

Step 7: Review Instance Launch

m5.xlarge	-	4	16	EBS only	Yes	Up to 10 Gigabit
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▼ Security Groups

[Edit security groups](#)

Security group name
Description

FPGA Developer AMI-1-10-0-AutogenByAWSMP-4
This security group was generated by AWS Marketplace and is based on recommended settings for FPGA Developer AMI version 1.10.0 provided by Amazon Web Services

Type ⓘ	Protocol ⓘ	Port Range ⓘ	Source ⓘ	Description ⓘ
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This security group has no rules

▶ Instance Details

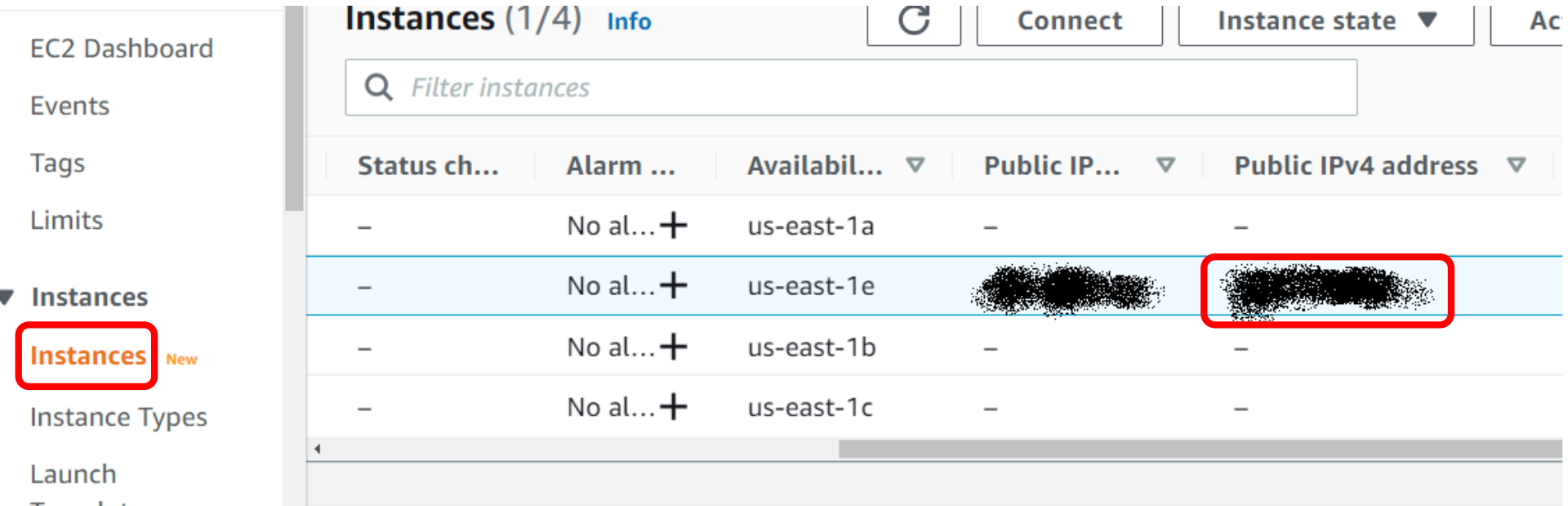
[Edit instance details](#)

▼ Storage

[Edit storage](#)

Volume Type ⓘ	Device ⓘ	Snapshot ⓘ	Size (GiB) ⓘ	Volume Type ⓘ	IOPS ⓘ	Throughput (MB/s) ⓘ	Delete on Termination ⓘ	Encrypted ⓘ
Root	/dev/sda1		120	gp2	360 / 3000	N/A	Yes	Not Encrypted
ebs	/dev/sdb		5	gp2	100 / 3000	N/A	No	Not Encrypted

Finding IP address



EC2 Dashboard

Events

Tags

Limits

Instances

Instances New

Instance Types

Launch

Instances (1/4) [Info](#)

Status ch...	Alarm ...	Availabil...	Public IP...	Public IPv4 address
–	No al...+	us-east-1a	–	–
–	No al...+	us-east-1e	[REDACTED]	[REDACTED]
–	No al...+	us-east-1b	–	–
–	No al...+	us-east-1c	–	–

Managing Instances

The screenshot displays the AWS Management Console's EC2 Instances page. The left sidebar shows navigation options: EC2 Dashboard, Events, Tags, Limits, Instances (selected), Instance Types, Launch, and Templates. The main content area shows a table of instances with columns for Name, Instance ID, and Instance state. One instance is selected, and its state is 'Running'. A dropdown menu is open, showing actions: Launch instances, Launch instance from template, Connect, Stop instance (highlighted with a red box), Start instance (highlighted with a green box), Reboot instance, Hibernate instance, Terminate instance (highlighted with a purple box), Instance settings, and Networking. A purple arrow points from the 'Terminate instance' option to the 'Volumes' link under the 'Elastic Block Store' section.

	Name	Instance ID	Instance state
☐	[Redacted]	[Redacted]	Stopped
☒	[Redacted]	[Redacted]	Running
☐	[Redacted]	[Redacted]	Stopped
☐	[Redacted]	[Redacted]	Stopped

▼ Elastic Block Store

Volumes

SSH into AWS Instance (Linux)

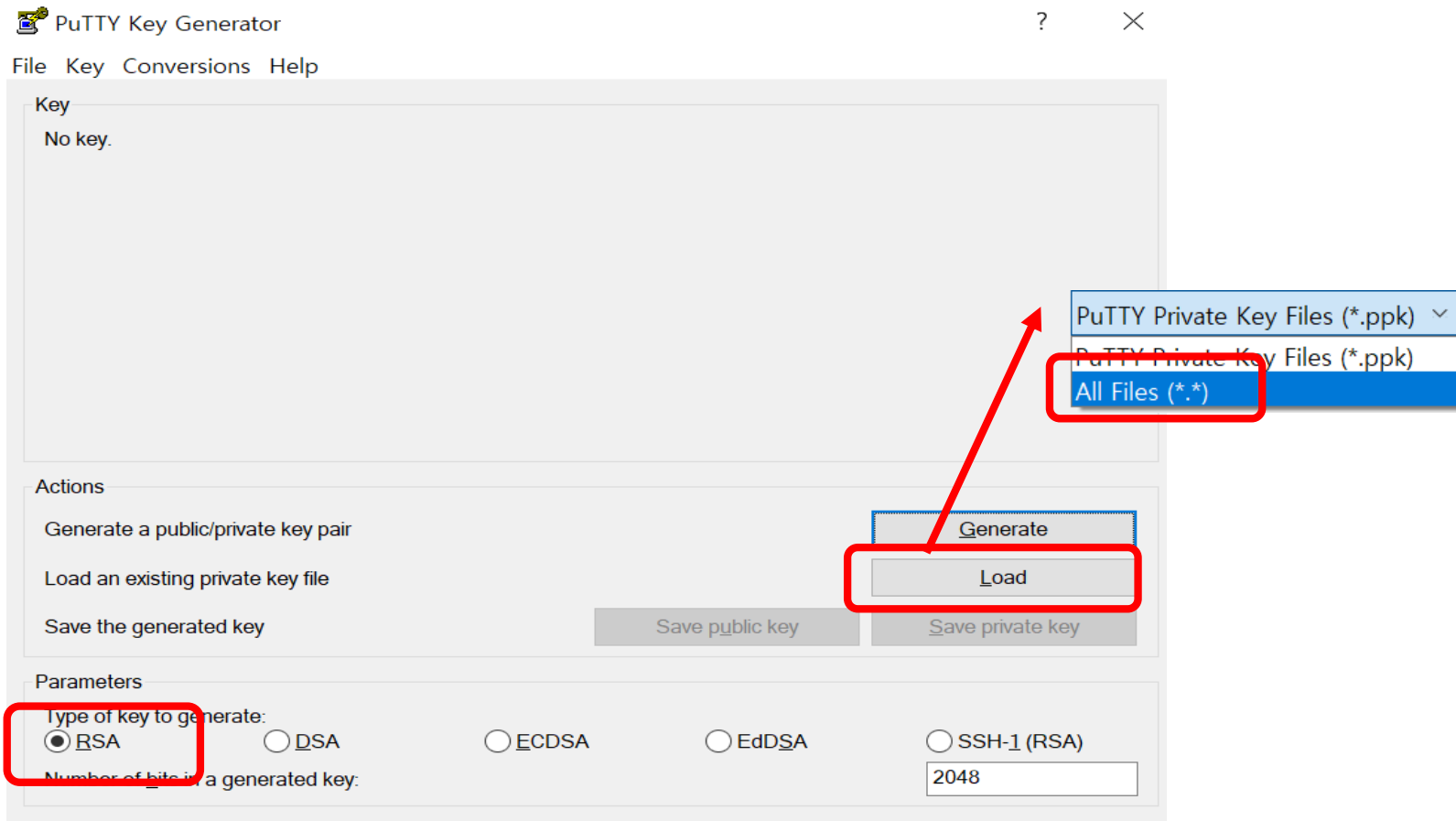
- You the following into your Linux console:
 - `ssh -i <your_keypair_file.pem> centos@<ip address>`
- If necessary, type
 - `chmod 400 <your_keypair_file.pem>`

SSH into AWS Instance (Windows)

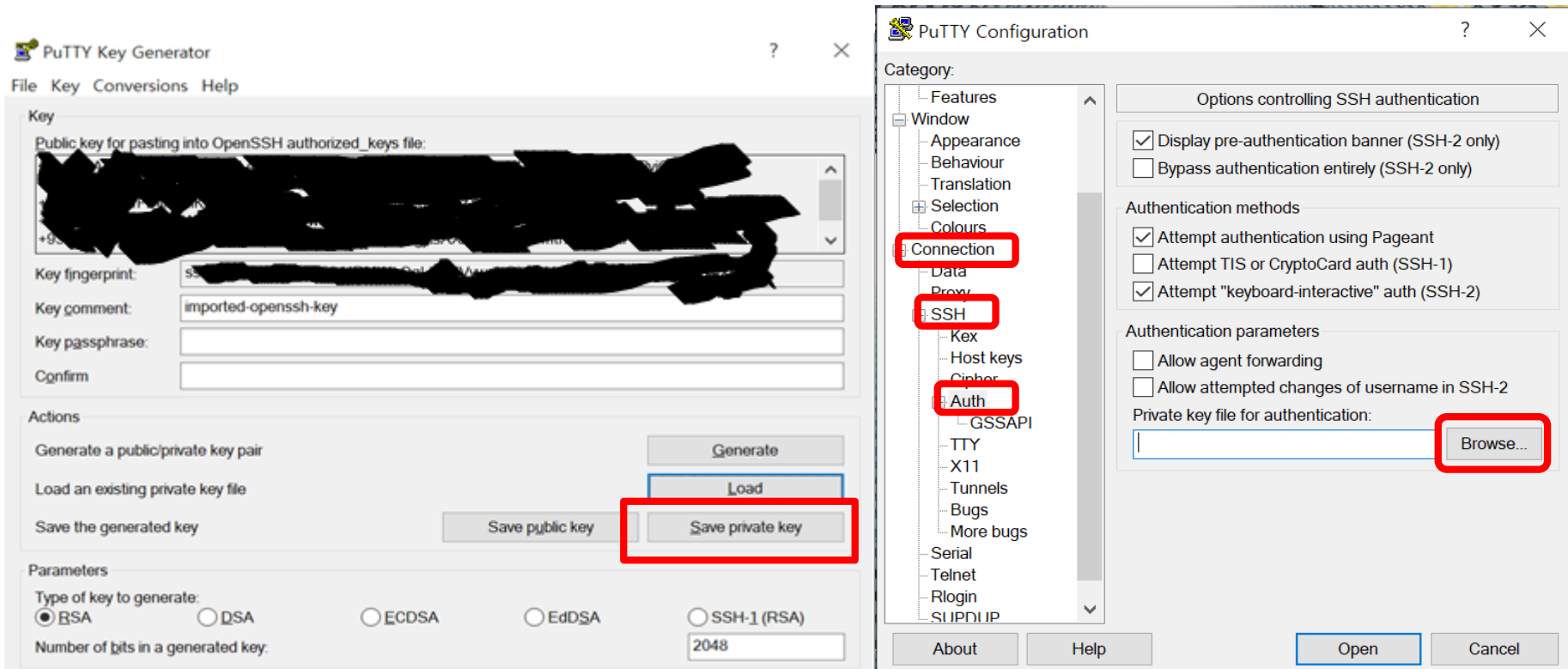
- You can use any SSH software
 - Example: Putty
 - Download from putty.org, and install software



Putty's Private Keygen (Windows)

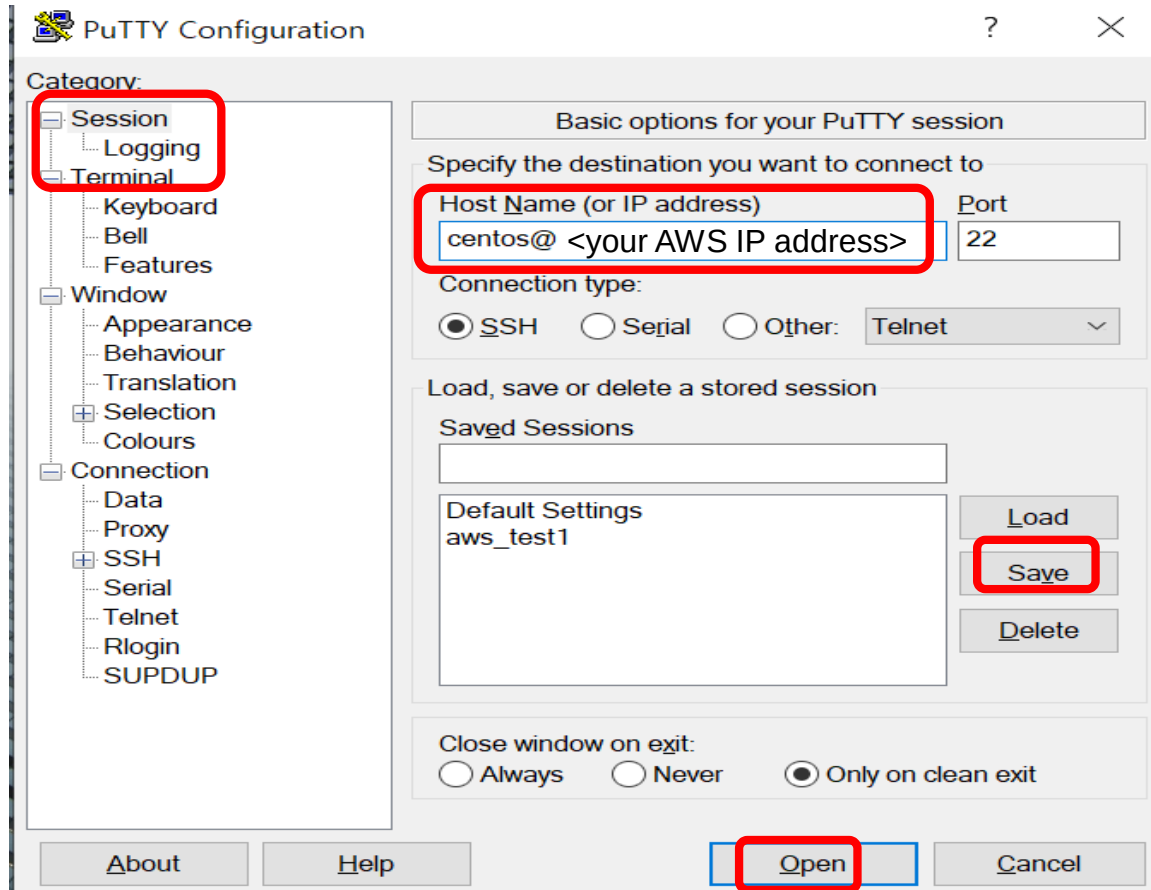


Putty's Private Keygen (Windows)



Saves to ppk file

SSH into AWS Instance (Windows)



F1 Instance Limit Increase

EC2 Dashboard

Events

Tags

Limits

Region: US East (Northern Virginia)

Primary Instance Type: All F instances

Limit: Instance Limit

New limit value: 8

Add another request

Limits

Calculate vCPU limit

Request limit increase

Search: F1

All limits

	Name	Limit type	Current limit
☐	Running INF1 Dedicated Hosts	Dedicated Hosts	0
☒	Running F1 Dedicated Hosts	Dedicated Hosts	0

to help resolve your case

since limits are based on the number of virtual central processing

and Instance limits, see our [Amazon EC2 On-Demand Instance L](#)

vice Quotas to view and manage your updated vCPU quotas. Lea

Use case description

Will be using F1 instance for class.

Summary

- In F1 instance, FPGA is connected to host CPU via PCIe interface
- Users can create an FPGA image on any EC2 instance using FPGA Developer AMI
- SSH into an AWS instance with a keypair
- Special attention is needed to stop (or terminate) an unused instance to avoid overbilling